Vector Quantization of Diffusion Transformers

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Abstract

This study examines vector weight quantization of diffusion transformers. One hypothesis is tested: calibrating codebooks by reconstructing the full-precision network output improves generation at fixed logical weight storage. Starting from a transferred, layerwise-calibrated VQDM representation of DiT-XL/2, only codebook values are optimized for 500 updates. Assignments, row scales, biases and other parameters remain fixed. In a matched ImageNet256 evaluation with 50 DDPM steps, guidance 1.5 and 10,000 images per model, FID decreases from 15.40 to 7.81 at 4.15 bits per DiT parameter. The full-precision reference reaches 6.10. This supports the hypothesis in the tested setting without establishing robustness across independent runs.

1 Introduction and goal

Diffusion transformers provide a useful generation model but require substantial weight storage [5]. Vector quantization replaces groups of weights with references to small codebooks. Errors left by independent layer calibration may interact through the complete network, motivating an additional optimization stage.

Goal. Reduce the generation-quality loss of a vector-quantized DiT without increasing its logical weight-storage budget.

Main idea. Calibrate the vector-quantized representation using reconstruction of the original denoiser's complete output. Jointly adjust codebook values while keeping assignments and other weights fixed. This is a calibration stage of weight quantization and the single idea examined in the article and its one-slide summary.

2 Related work

DiT uses a transformer denoiser on latent image patches [5]. VQDM combines vector weight quantization with global output-reconstruction fine-tuning for SDXL and SDXL-Turbo [3]. Here this calibration stage is transferred to DiT with only codebook values updated. The control contains the transferred layerwise stage without subsequent global fine-tuning, rather than the complete original VQDM pipeline.

Output reconstruction is the calibration objective used here. Since its targets are produced by a frozen full-precision network, this particular objective also supplies a knowledge-distillation signal [4], as discussed in VQDM. Quantization identifies the compact weight representation and calibration procedure; distillation describes the source of the training target. Reconstruction losses can also be used at layer or block level. VQ4DiT calibrates codebooks and candidate assignments through transformer-block outputs [1]. It provides context but is not a reproduced baseline here. Quality measurements use the official ADM toolkit [2].

Preprint.

3 Problem statement

Hypothesis. For the same layerwise-quantized initialization, architecture and storage precision, calibrating only codebook values with a full-network output-reconstruction loss lowers generation FID while preserving logical storage.

Model. The teacher is the public class-conditional DiT-XL/2 checkpoint for ImageNet 256×256. The student has the same architecture. Five matrices per block are quantized: attention QKV and output projections, both MLP projections, and adaLN modulation. The 28 blocks contain 140 such matrices and 668,860,416 quantized weights. Activations, biases and parameters outside these matrices are not quantized.

A length-d weight subvector is reconstructed as

$$\widehat{w}_{\ell,r,g} = s_{\ell,r} \sum_{m=1}^{M} C_{\ell,m}[a_{\ell,r,g,m}], \qquad M = 4,\ K = 256,\ d = 8. \tag{1}$$

Here ℓ denotes a layer, r an output row, g a subvector, C codebooks, a stored indices and s row scales. Indices require nominally $M \log_2 K/d = 4$ bits per quantized weight. Including codebooks, scales and unchanged parameters gives 4.150078 logical bits per whole-DiT parameter, excluding the VAE.

Let $f_\theta(z_t, t, y) = (\epsilon_\theta, v_\theta)$ be the teacher and $f_{Q(C)}$ the student. The second head v is the raw learned-variance prediction, not the variance itself. For a real-image VAE latent z_0, inputs are sampled using

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t}\, \eta, \qquad \eta \sim \mathcal{N}(0, I), \quad t \sim \mathrm{Uniform}\{0, \ldots, 999\}. \tag{2}$$

Both networks receive the same (z_t, t, y). The train/reserved split is fixed. This global calibration stage uses real-image latents and is not data-free.

Quality criteria. The primary criterion is lower FID on a matched 10,000-image evaluation, subject to unchanged logical bpp. FID compares means and covariances of generated and reference Inception features. Lower sFID is an auxiliary spatial-feature criterion; higher IS, precision and recall are also reported. Precision and recall describe different aspects of sample fidelity and distribution coverage. A reduction of teacher-output MSE is a separate diagnostic, not a substitute for generation metrics. The experiment tests an exploratory hypothesis in one fixed configuration, not a statistical claim across training seeds.

4 Solution: properties and algorithm

Training objective. Codebooks minimize

$$\mathcal{L}_{\mathrm{rec}} = \mathrm{MSE}(\epsilon_{Q(C)}, \epsilon_\theta) + 0.1\, \mathrm{MSE}(v_{Q(C)}, v_\theta). \tag{3}$$

Teacher parameters, assignments, row scales, biases and other weights remain fixed. The target is the teacher's prediction, rather than the sampled noise η. The variance term is regression on the teacher head, not a DDPM variational-bound loss. Classifier-free guidance is not applied in the training objective; labels are replaced by the null class with probability 0.1.

Property 1: unchanged logical storage. For matrices of size $o_\ell \times i_\ell$, a fixed representation has storage

$$B_{\mathrm{bits}} = \sum_{\ell} \frac{o_\ell i_\ell}{d} M \lceil \log_2 K \rceil + 16 \sum_{\ell} MKd + B_{\mathrm{fixed}}. \tag{4}$$

The first term stores indices, the second FP16 codebooks, and B_{fixed} contains unchanged scales and other parameters. None of these array sizes or precisions depends on the updated codebook values. Therefore codebook-only optimization preserves logical bpp. This is a counting argument for the packed representation, not a bound on serialized checkpoints or inference time.

Property 2: a prediction-error bound. Write $S = \epsilon_{Q(C)}$, $T = \epsilon_\theta$ and let D be the number of noise-output elements. On the common input distribution, assume finite second moments and define

$$R(S) = \mathbb{E}\|S - \eta\|^2/D, \qquad D_{\mathrm{rec}} = \mathbb{E}\|S - T\|^2/D. \tag{5}$$

The triangle inequality in L^2 gives

$$\sqrt{R(S)} \le \sqrt{D_{\mathrm{rec}}} + \sqrt{R(T)}. \tag{6}$$

Indeed, $S - \eta = (S - T) + (T - \eta)$. Output reconstruction controls one term in this upper bound; it does not guarantee a reduction of actual noise error or FID. The latter is checked by generation.

Algorithm.

1. Load the completed layerwise VQDM student and its full-precision teacher. Freeze all student parameters except codebook master values.
2. For each of 500 updates, sample a batch of 16 training latents, a time and Gaussian noise, and construct inputs by Equation (2). Apply null-class replacement with probability 0.1.
3. Evaluate teacher and student on identical inputs. Compute Equation (3), backpropagate through the entire student, and update codebooks using Adam with learning rate 10^{-5}.
4. Monitor a fixed held-out teacher-noise probe. Export FP16 codebooks at step 500 and check that assignments and frozen parameters are unchanged. The final500 checkpoint is fixed before the 10,000-image FID calculation.
5. Generate matched image sets for the full-precision reference, layerwise-only model and globally calibrated model, then evaluate the same external criteria for all three.

FP32 master values are optimized with autograd through FP16 casts; reconstructed weights are used in FP32 computation. The implementation's cast gradients are used, without claiming exact differentiation of a discrete rounding map. Gradients are not unrolled through the complete sampling trajectory.

5 Experiment

Objectives. Test whether network-output calibration lowers generation FID from the same vector-quantized initialization, verify the storage invariant, and examine auxiliary metrics and the remaining gap to the reference. The full-precision model is a quality reference; the layerwise-only quantized model is the direct control.

Calibration and training data. Initial layerwise calibration used 256 teacher-generated DDPM trajectories. Global codebook calibration uses 5,000 fixed ImageNet training images selected from two fixed shards and encoded by the MSE VAE: 4,000 latents for updates and 1,000 reserved. A fixed probe of 96 input/noise pairs at six timesteps is drawn from the reserved subset. Its MSE is not an average over all 1,000 reserved images. The saved reference-prediction probe is monitored at updates 0, 50, 100, 200 and 500. Final500 also has the lowest saved probe error, without selection by FID.

Evaluation protocol. Each model generates 10,000 images, ten per ImageNet class, using common sample records and paired initial/per-step noise, with selection seed 20261103. Generation uses **DDPM50**, CFG 1.5, `sd-vae-ft-mse`, FP32 computation, disabled TF32 and mathematical attention. The 1,000-step training noising schedule is distinct from these 50 generation steps.

The official ADM Inception graph and ImageNet256 reference are fixed. FID/sFID use stored full-dataset reference statistics; precision/recall use its 10,000 reference images. Each archive extends an earlier same-protocol 5,000-image prefix. All reported metrics are recomputed on the full 10,000-image sets. No result on 5,000 images is compared with a 10,000-image control.

6 Results and error analysis

Table 1: Model–data–criteria summary. IN256/10k denotes ImageNet 256×256, 10,000 generated images per model. All rows use DDPM50, CFG 1.5 and the same MSE VAE. Logical bpp excludes the VAE.

Model	Evaluation data	bpp	FID↓	sFID↓
FP reference	IN256 / 10k	32	6.098	21.675
Layerwise VQDM	IN256 / 10k	4.1501	15.402	27.601
VQDM + output calibration	IN256 / 10k	4.1501	7.814	23.367

Table 2: Auxiliary quality criteria on the same ImageNet256/10k sets as Table 1. Higher values are preferred.

Model	IS↑	Precision↑	Recall↑
FP reference	253.827	0.7904	0.6659
Layerwise VQDM	145.312	0.6491	0.7115
VQDM + output calibration	211.549	0.7427	0.6770

Effect and remaining error. FID decreases from 15.40 to 7.81, a relative reduction of 49.3%, while logical storage remains 4.15 bpp. The saved check reports zero changes among 334,430,208 assignments. sFID, IS and precision improve, but recall falls from 0.7115 to 0.6770. The result therefore includes a coverage trade-off. The calibrated model's FID remains 1.72 above its full-precision reference.

The held-out teacher-noise MSE decreases from 0.001162851 to 0.001007803, or 13.33%. This same-input prediction error and the generation FID quantify different effects. Their improvements are reported separately.

Measurement and uncertainty. Independent replay with a freshly fetched official ADM CLI reproduces checked FIDs within 10^{-5}, including a published fixed ADM fixture. Some recall values differ at the fourth decimal place. This validates the checked FID calculation, not bitwise identity of every metric or reproduction of another paper's compression results.

Only one fixed quantizer initialization and optimization setting is considered. No confidence interval across independently trained models is estimated, and a single generated cohort is reused exploratorily. Training data comes from two fixed shards; class-balanced evaluation differs from natural ImageNet frequencies. There is no compute-matched alternative fine-tuning control, so the result demonstrates the benefit of adding global output-reconstruction calibration without establishing its superiority to every other use of 500 updates.

7 Conclusion

The hypothesis is supported in the tested configuration: calibrating vector-quantization codebooks by network-output reconstruction lowers FID from 15.40 to 7.81 at unchanged logical storage. A residual gap to FP and a recall decrease remain. The completed experiment evaluates a calibration choice within vector quantization of DiT. Independent repeats and a compute-matched objective comparison are needed before a broader conclusion.

Artifacts. The manuscript, one-slide summary, three-minute talk and numeric snapshot are available in the practicum repository. Scientific code and experiment history remain in the research repository. This is an intermediate practicum manuscript.

References

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